Variational Autoencoders on Two-Moons Data and Conditional MNIST

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Abstract

This report studies variational autoencoders (VAEs) on a synthetic two-dimensional two-moons distribution and on handwritten MNIST digits. The experiments vary both the dimension of the latent space and the width/depth of the neural network. Performance is evaluated using reconstruction and Kullback–Leibler (KL) losses, but generated samples are treated as the primary qualitative criterion. On the two-moons problem, a latent dimension of two is sufficient to represent the distribution, and increasing the latent dimension does not improve the samples. On conditional MNIST, the two-dimensional latent model produced the most convincing samples in the visual comparison, even though larger latent spaces achieved lower reconstruction losses. Changing network width and depth had a smaller visual effect than changing the latent dimension; the deepest model was only marginally better than the medium and shallow alternatives.

1 Introduction

Variational autoencoders combine an encoder, a probabilistic latent representation, and a decoder. Unlike a conventional autoencoder, a VAE is trained so that its latent representation remains close to a chosen prior, usually a standard normal distribution. This makes it possible to generate new samples by drawing a latent vector from the prior and passing it through the decoder.

The goal of this assignment was to implement a VAE, first on an easy-to-visualize two-dimensional distribution and then on a more interesting conditional image generation problem. The experiments address two questions:

1. How does the size of the latent space affect reconstruction, latent regularization, and generated samples?
2. How much does network width and depth matter when the latent dimension is held fixed?

The analysis emphasizes newly generated samples rather than relying only on training or test error. This distinction is important because a model can reconstruct held-out examples well without producing useful samples from its prior.

2 Model and objective

2.1 VAE formulation

For an input x, the encoder produces a mean $\mu(x)$ and log variance $\log \sigma^2(x)$. A latent sample is obtained with the reparameterization trick:

$$z = \mu(x) + \sigma(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

The decoder maps z to reconstruction logits. For a conditional model, the class label is converted to a one-hot vector and concatenated to the encoder input and decoder latent vector. Thus, the class label is available both when encoding an image and when generating a new image.

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{reconstruction}} + \beta D_{\text{KL}} \left(q_\phi(z \mid x) \,\|\, p(z) \right),$$

where

$$D_{\text{KL}} = -\frac{1}{2} \sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2 \right).$$

The two-moons experiments use mean-squared error for reconstruction and $\beta_{\max} = 0.25$. MNIST uses binary cross-entropy with logits and $\beta_{\max} = 1$. In both cases, β is warmed up over the first half of training. The reported reconstruction and KL values are separate components; KL is a regularization term, not another reconstruction error.

2.2 Implementation and data

The implementation uses PyTorch and the Adam optimizer with learning rate 10^{-3}. The full experiments use 40 epochs and a fixed random seed of 42. The two-moons data contain 4,000 noisy points, split into 70% training, 15% validation, and 15% test data. The points are standardized using statistics computed from the training split.

For MNIST, the conditional VAE receives flattened 28×28 grayscale images and one of ten digit labels. The experiment uses 12,000 training examples, 2,000 validation examples, and 2,000 held-out test examples. The default latent-dimension architecture has hidden layers $(256, 128)$. Generated images are obtained by sampling $z \sim \mathcal{N}(0, I)$ and conditioning the decoder on a requested digit label.

3 Two-moons experiments

3.1 Latent dimension

The first experiment keeps the hidden architecture fixed at $(64, 64)$ and compares latent dimensions 1, 2, 4, and 8. The generated samples are shown in Figure 1. A one-dimensional latent space can produce a rough approximation, but a two-dimensional latent space is sufficient to capture the two curved components. Increasing the dimension beyond two does not visibly improve the distribution and can make the generated points less focused.

Table 1: Two-moons latent-dimension results.

Latent dimension	Parameters	Validation total	Test reconstruction	Test KL
1	8,900	0.659	0.260	1.730
2	9,094	0.680	0.222	1.907
4	9,482	0.698	0.218	1.915
8	10,258	0.706	0.230	1.875

The numerical results do not identify a single winner independently of the samples. Latent dimension four has the lowest test reconstruction loss by a small margin, while latent dimension two has a slightly better validation total than dimensions four and eight. The visual result is

more informative for this low-dimensional task: two dimensions provide enough capacity without introducing unnecessary degrees of freedom.

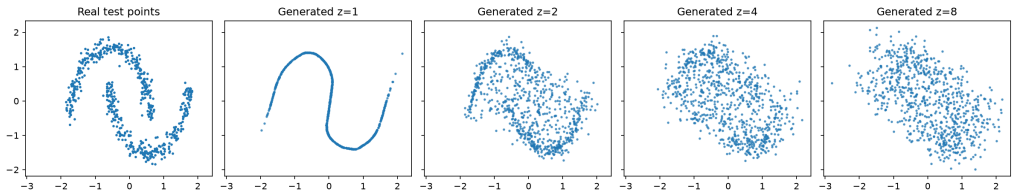


Figure 1: Two-moons data and newly generated samples for different latent dimensions. The samples should be judged by whether they form the two curved components rather than by pointwise reconstruction alone.

3.2 Network width and depth

The second two-moons experiment fixes the latent dimension at two and changes the hidden layers. The shallow model uses one layer of width 32, the medium model uses $(64, 64)$, and the deep model uses $(128, 128, 128)$.

Table 2: Two-moons architecture results with latent dimension two.

Architecture	Parameters	Validation total	Test reconstruction	Test KL
Shallow-32	390	0.704	0.248	1.869
Medium-64x2	9,094	0.680	0.222	1.907
Deep-128x3	67,590	0.684	0.198	1.950

The deep model has the lowest test reconstruction loss, but it uses more than seven times as many parameters as the medium model and more than 170 times as many as the shallow model. The generated samples are all broadly successful, so the visual improvement from adding depth and width is modest compared with the computational cost. This supports the conclusion that the latent dimension is the more important design choice for this distribution.

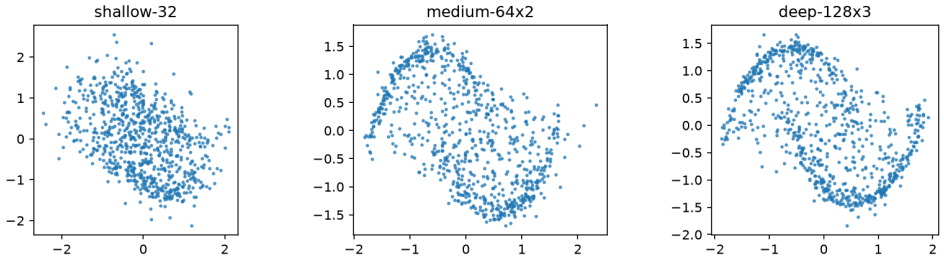


Figure 2: Generated two-moons samples for the shallow, medium, and deep architectures.

4 Conditional MNIST experiments

4.1 Latent dimension

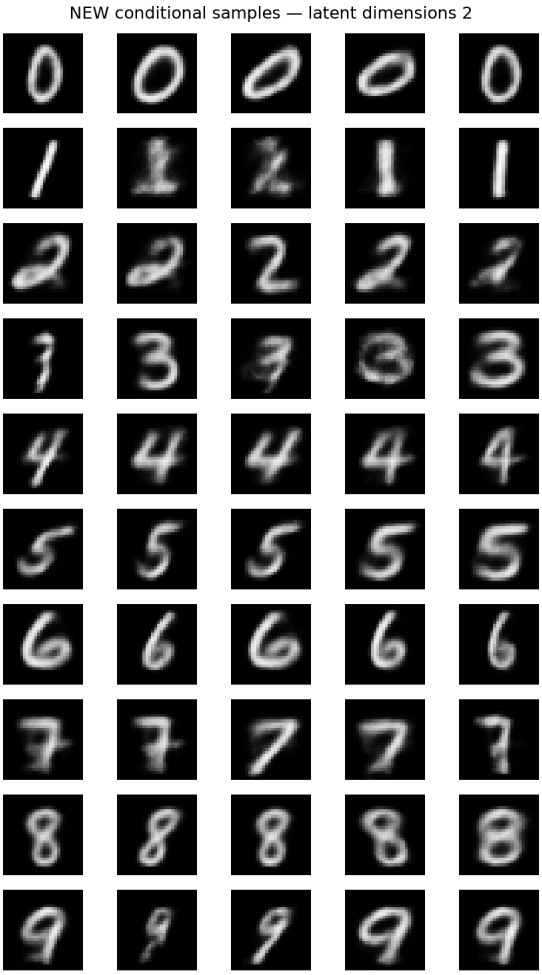
The MNIST latent-dimension experiment uses the same conditional architecture for latent dimensions 2, 4, 8, 16, and 32. The generated grids request digits 0 through 9 from the decoder. The results are summarized in Table 3, and representative grids are shown in Figure 3.

Table 3: Conditional MNIST latent-dimension results.

Latent dimension	Parameters	Validation total	Test reconstruction	Test KL
2	473,108	134.937	130.714	4.772
4	473,880	120.546	113.112	8.011
8	475,424	109.528	95.071	14.085
16	478,512	106.615	86.834	18.927
32	484,688	106.386	86.899	19.009

The numerical reconstruction loss decreases substantially as the latent dimension increases, with most of the improvement occurring by dimension 16. The KL term increases because the larger latent representations carry more information and therefore deviate more from the unit Gaussian prior. This is not evidence that the larger models are incorrect: a lower KL term is not always desirable, since excessive regularization can remove information needed to represent the data.

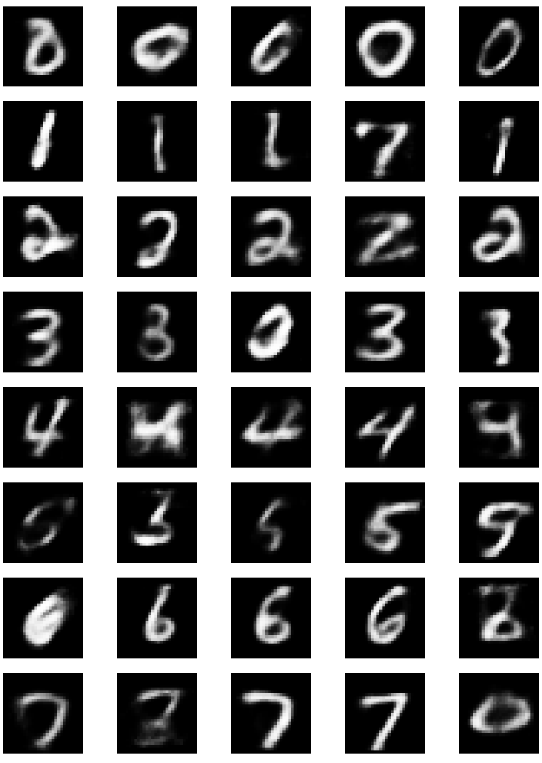
The visual comparison gives a different result. In the generated grids, the two-dimensional model produces the most recognizable and consistent class-conditioned digits in this run. The larger latent models have lower reconstruction losses, but their prior samples are visually less convincing and progressively less reliable at following the requested digit. This illustrates why generated samples must be examined in addition to reconstruction metrics. It also suggests that a larger latent space can make the posterior represent training examples more accurately without making the prior-to-decoder path easier to sample from after finite training.

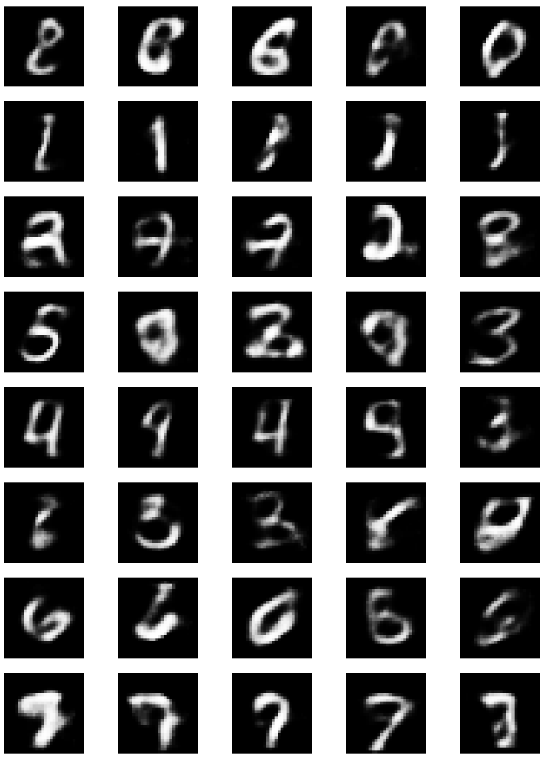

NEW conditional samples — latent dimensions 2

(a) $z = 2$


NEW conditional samples — latent dimensions 4

(b) $z = 4$


NEW conditional samples — latent dimensions 8


NEW conditional samples — latent dimensions 16

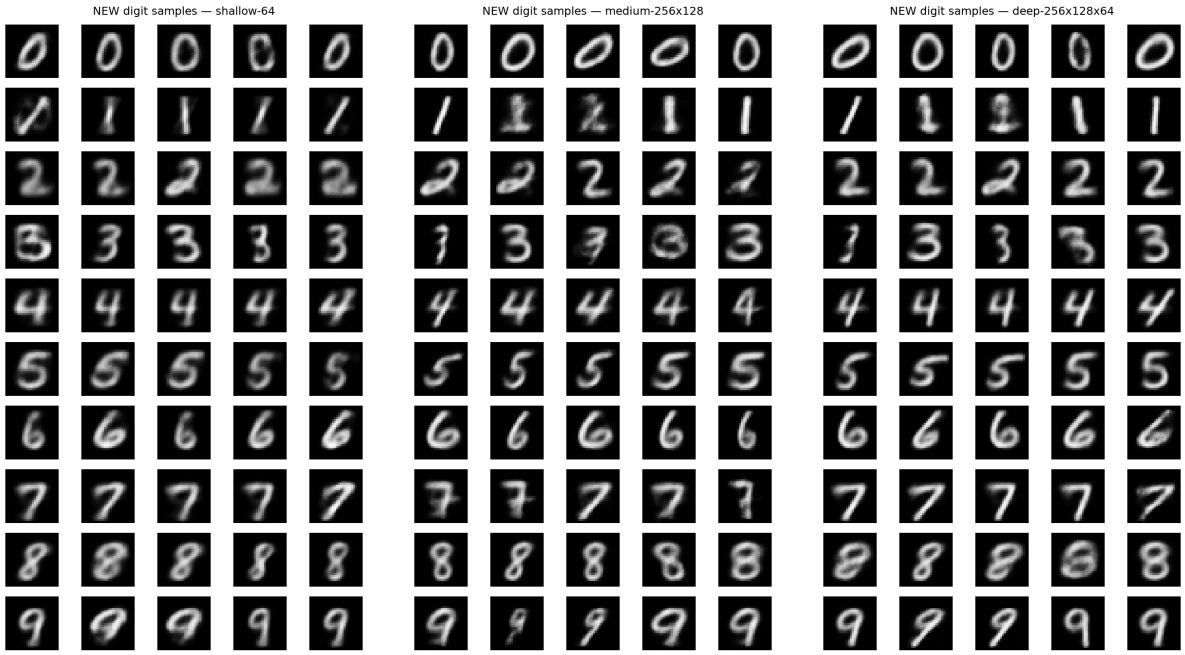
4.2 Network architecture

The architecture experiment fixes the latent dimension at two and compares three conditional networks. The shallow, medium, and deep models use hidden layers (64), (256, 128), and (256, 128, 64), respectively.

Table 4: Conditional MNIST architecture results with latent dimension two.

Architecture	Parameters	Validation total	Test reconstruction	Test KL
Shallow-64	102,932	143.882	139.628	3.963
Medium-256x128	473,108	134.937	130.714	4.772
Deep-256x128x64	488,596	134.004	130.245	4.559

The deep model has the best reported validation total and the lowest test reconstruction error, but its advantage over the medium model is small. The generated images also look broadly similar across the three architectures. The shallow network has fewer parameters and performs worse numerically, but the medium and deep models are close in both sample quality and metrics. Thus, for this experiment, increasing network depth and width has less visible impact than changing the latent dimension.



(a) Shallow (b) Medium (c) Deep

Figure 4: New conditional MNIST samples for the architecture comparison.

4.3 Reconstructions versus new samples

The final visualization separates three different tasks: reproducing held-out images, generating new images from the prior, and obeying the requested condition. Reconstructions pass test images through the encoder and decoder, whereas new samples draw random latent vectors and provide only a digit label to the decoder. The latter is the more demanding generative test.

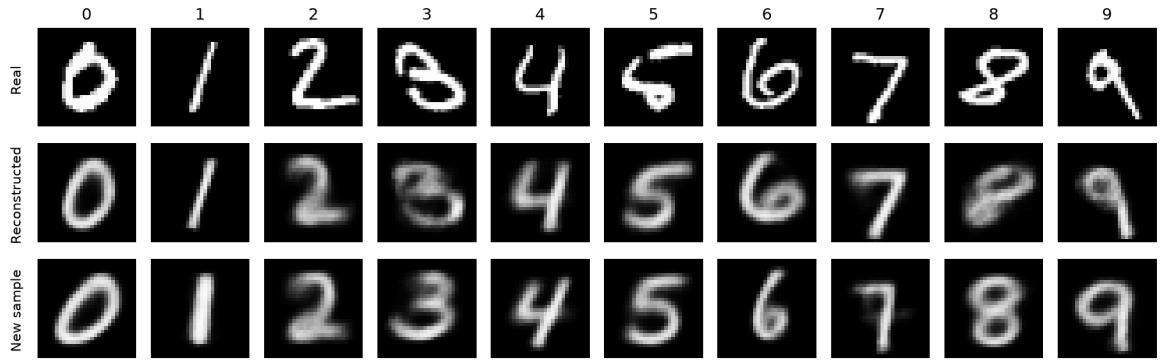


Figure 5: Real test examples, their reconstructions, and newly generated conditional samples. Column titles identify the requested digit.

5 Discussion

Several conclusions emerge from the experiments.

Latent size is not simply a capacity knob. Increasing the latent dimension reduced MNIST reconstruction error, but the visual quality of prior samples did not improve in the same way. The decoder is trained primarily on latent vectors produced by the encoder. Sampling from the prior requires the aggregate encoded distribution to align sufficiently with that prior. A larger latent space can therefore improve reconstruction while making prior sampling harder under the same training budget.

KL should be interpreted jointly with reconstruction. The KL term measures the distance between the encoder's approximate posterior and the standard normal prior. A small KL can indicate strong regularization, but it can also mean that the latent code carries little information. A large KL can accompany better reconstructions because the model is using the latent variables more actively. The relevant objective is the weighted sum, together with the quality and diversity of generated samples.

Architecture has diminishing returns. For both datasets, larger networks reduce reconstruction error to some degree. However, the visual difference between medium and deep networks is small, especially on MNIST with a two-dimensional latent space. The additional parameters are not automatically converted into better generative behavior. This result is consistent with the idea that the bottleneck and prior regularization constrain the model more strongly than the decoder's extra capacity in these experiments.

Evaluation should include generation. The two-moons plots make distributional errors immediately visible, while the MNIST grids reveal whether the requested class is preserved. These checks would be missed by reporting only a reconstruction loss. In particular, the MNIST latent-dimension experiment demonstrates that the numerically best reconstruction model is not necessarily the visually best model for new conditional samples.

6 Limitations and future work

The experiments use one random seed, a subset of MNIST, and a fixed training schedule. The full MNIST runs use 40 epochs, so the apparent degradation of larger latent spaces could partly reflect optimization time or the chosen KL warm-up schedule rather than an inherent limitation of those dimensions. The visual ranking is also based on a finite number of generated images and human judgment. A more complete study would repeat each configuration across several seeds, use a larger training set, tune the warm-up schedule, and evaluate conditional sample quality with an automated classifier or a class-conditional likelihood metric. It would also be useful to compare latent dimensions and architectures using the same selected validation criterion before reporting the held-out test results.

7 Conclusion

The VAE successfully learned the two-moons distribution and generated class-conditioned MNIST digits. On the two-moons problem, a latent dimension of two was sufficient; larger latent spaces did not provide a meaningful visual benefit. On MNIST, larger latent spaces produced lower reconstruction losses, but the two-dimensional model produced the best-looking new conditional samples in the visual comparison. With latent dimension fixed at two, the deep architecture was numerically and visually only slightly better than the medium model, while the shallow model used far fewer parameters but lost some performance. Overall, these results support selecting a VAE based on both reconstruction/regularization metrics and the quality of samples drawn from the latent prior.